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SUPPORT PLATE FOR FOLDING PORTABLE DISPLAY DEVICE

Abstract

An example folding device comprising: a first housing; a second housing; a hinge assembly rotatably connected to the first housing and the second housing; a continuous display connected to and across the first housing and the second housing; and a first collapsible support plate connected to the first housing and configured to support the continuous display when the folding device is fully opened, wherein: the first collapsible support plate comprises a single continuous plate folded over itself to form two parallel plates, a first parallel plate of the two parallel plates defines a straight edge that overhangs one or more voids in a second parallel plate of the two parallel plates, and the straight edge defined by the first parallel plate is adjacent to the hinge assembly.

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Background/Summary

BACKGROUND

[0001] Devices that include displays may be referred to as display devices. In general, it may be desirable to increase a size of a display (e.g., the area on which images are displayed) as much as possible. Increasing the size of a display may make the device that includes the display large and unwieldy. For instance, devices with larger displays may not fit in pockets, bags, and the like. One way to increase the size of a display without negatively affecting the portability of the device is to make the device collapsible such that the display can be folded (e.g., in half).

SUMMARY

[0002] In general, aspects of this disclosure are directed to supporting plates (i.e., support plates) of a folding device. Folding devices may generally include two housings connected via a hinge assembly and a continuous display that spans the two housings. Folding devices may include a support plate, comprising two separate plates, that support a rear of the continuous display when the folding device is fully opened. Such designs may assist in preventing or limiting deformation in the continuous display and/or provide consistent support of the continuous display when the folding device is open (e.g., as a user drags their finger across the span of the continuous display between the two housings). However, such designs may not be desirable. For instance, the use of two separate plates to form the support plate may occupy additional space within the folding device, increase the weight of the folding device, and/or increase a manufacturing cost of the folding device.

[0003] In accordance with one or more aspects of this disclosure, a folding device may include a single continuous plate folded over itself to form two parallel plates that operate as a support plate for the folding device. For instance, the single continuous plate may include a fold to form a first plate, parallel to, a second plate. The second plate may comprise assembly geometry and weight reduction geometry, while the first plate may comprise a clean surface (i.e., limited assembly geometry and no weight reduction geometry). The first plate may include a straight edge, opposite the fold, that overhangs a non-straight and complex outline of the second support plate, without inhibiting movement of hinge assembly components. As such, when the folding device is open, deformations in the continuous display are mitigated (e.g., any distortions or changes in the continuous display caused by the non-straight and complex outline of the second plate may be prevented or limited). In this way, aspects of this disclosure may reduce manufacturing costs, weight, and/or occupation of space within the folding device, while preventing or limiting deformation in the continuous display and/or providing consistent support of the continuous display when the folding device is open.

[0004] In one example, a folding device includes a first housing; a second housing; a hinge assembly rotatably connected to the first housing and the second housing; a continuous display connected to and across the first housing and the second housing; and a first collapsible support plate connected to the first housing and configured to support the continuous display when the folding device is fully opened, wherein: the first collapsible support plate comprises a single continuous plate folded over itself to form two parallel plates, a first parallel plate of the two parallel plates defines a straight edge that overhangs one or more voids in a second parallel plate of the two parallel plates, and the straight edge defined by the first parallel plate is adjacent to the hinge assembly.

[0005] The details of one or more examples of the disclosure are set forth in the accompanying drawings and the description below. Other features, objects, and advantages of the disclosure will be apparent from the description and drawings, and from the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIGS. **1A** and **1B** are schematic diagrams illustrating a folding device with a flexible continuous display, in accordance with one or more aspects of this disclosure.

[0007] FIG. **1C** is a conceptual diagram illustrating a cross-sectional view of a folding device with a flexible continuous display and example supporting plates, in accordance with one or more aspects of the disclosure.

[0008] FIGS. **2A** and **2B** are conceptual diagrams illustrating example hinge assemblies with supporting plates, in accordance with one or more aspects of this disclosure.

[0009] FIGS. **3A**, **3B**, and **3C** are conceptual diagrams illustrating example supporting plates, in accordance with one or more aspects of this disclosure.

[0010] FIGS. **4A** and **4B** are conceptual diagrams further illustrating example supporting plates, in accordance with one or more aspects of this disclosure.

[0011] FIG. **5** is a conceptual diagram illustrating a cross-sectional view of example supporting plate, in accordance with one or more aspects of this disclosure.

DETAILED DESCRIPTION

[0012] In general, this disclosure is directed to folding devices that include foldable continuous displays with a span supported by support plates (i.e., supporting plates). A folding device may include at least two housings (i.e., panels) and a hinge assembly configured to allow the housings to be moved into a collapsed state in which the device is considered closed and an expanded state in which the device is considered open. When the device is in the expanded state, a display may be visible and may cover at least a portion of an inner surface of the housings. As such, the device may be considered to be a continuous display (i.e., because it continues across or spans a boundary between the housings). By utilizing a single continuous plate to create the support plate, as opposed to two separate plates, the continuous display may be supported while reducing the weight, manufacturing costs, and/or occupation of space within the folding device. In this way, foldable continuous display devices may be improved.

[0013] FIGS. **1A** and **1B** are schematic diagrams illustrating a folding device **100** with a flexible continuous display **106**, in accordance with one or more aspects of this disclosure. Examples of device **100** include foldable mobile computing devices such as foldable smart phones, foldable tablets, foldable e-readers, foldable gaming systems, or any other foldable portable device that includes a display.

[0014] As shown in FIGS. **1A** and **1B**, device **100** includes a first housing **102**, second housing **104**, continuous display **106**, and hinge assembly **122**. Each of first housing **102** and second housing **104** may include an inner surface and an outer surface. When device **100** is closed the outer surface of first housing **102** may be visible when looking down at device **100** in the z-axis and the outer surface of second housing **104** may be visible when looking up at device **100** in the z-axis. The inner surfaces of first housing **102** and second housing **104** may not be externally visible when device **100** is closed. As further shown in FIG. **1A**, when folding device **100** is fully open, an inner surface of a first housing **102** is coplanar with an inner surface of a second housing **104**.

[0015] Computing and/or electrical components of device **100** may be distributed amongst first housing **102** and second housing **104**. For example, first housing **102** may include a main logic board and second housing **104** may include a battery. This is merely one example arrangement of components amongst first housing **102** and/or second housing **104**; other arrangements are possible. For instance, both first housing **102** and second housing **104** may include respective batteries.

[0016] Continuous display **106** may be capable of rendering data into images viewable by a user of device **100**. For example, continuous display **106** may include a matrix of pixels that are individually controllable. Examples of continuous display **106** include, but are not limited to, liquid crystal displays (LCD), light emitting diode (LED) displays, organic light-emitting diode (OLED)

displays, micro light-emitting diode (microLED) displays, or similar monochrome or color displays capable of outputting visible information to a user of device **100**. Continuous display **106** may span hinge assembly **122** from first housing **102** to second housing **104**.

[0017] In some examples, device **100** may include one or more displays in addition to continuous display **106**. For instance, device **100** may include a first additional display on the outer surface of first housing **102**. In some examples, device **100** may further include a second additional display on the outer surface of second housing **104**.

[0018] One or more of continuous display **106**, the first additional display, and/or the second additional display may be presence-sensitive displays. In some examples, a presence sensitive display may detect an object at and/or near a screen. As one example range, a presence-sensitive display may detect an object, such as a finger or stylus that is within 2 inches of the screen. The presence-sensitive display may determine a location (e.g., an (x,y) coordinate) of a screen at which the object was detected. In another example range, a presence-sensitive display may detect an object within six inches of the screen. Other ranges are also possible. The presence-sensitive display may determine the location of the screen selected by a user's finger using capacitive, inductive, and/or optical recognition techniques. In some examples, the presence sensitive display also provides output to a user using tactile, audio, or video stimuli.

[0019] In general, hinge assembly **122** may include supporting plates for support of a backside of the continuous display, such as continuous display **106**. The resulting support on continuous display **106** may assist in preventing or limiting deformation in continuous display **106** and/or provide a consistent support of continuous display **106** as a user drags their finger across the span of continuous display **106** between the first and second housing. In some examples, two separate plates may be used to form the supporting plates. Such a design may be undesirable. For instance, the use of two separate plates to create a support plate may result in additional space being occupied within the folding device **100**, an increase in the weight of the folding device **100**, and/or an increase in a manufacturing cost of the folding device **100**.

[0020] In accordance with one or more aspects of this disclosure, a single continuous plate folded over itself to form two parallel plates may be utilized as a support plate for the folding device **100**. For instance, the single continuous plate may include a fold (i.e., a bend, a crease, etc.) to transform the single continuous place into a first plate, parallel to and still connected to, a second plate. The second plate may comprise assembly geometry and weight reduction features, while the first plate may comprise a clean surface (i.e., limited assembly geometry and no weight reduction features). The first plate may include a straight edge, opposite the fold, that overhangs a non-straight and complex outline of the second support plate, without inhibiting movement of hinge assembly components. As such, when the folding device **100** is open deformations in the continuous display are mitigated (i.e., any distortions or changes in the continuous display caused by the non-straight and complex outline of the second plate may be prevented or limited). In this way, aspects of this disclosure may reduce manufacturing costs, weight, and/or occupation of space within the folding device, while preventing or limiting deformation in the continuous display and/or providing consistent support of the continuous display when the folding device is open.

[0021] FIG. **1C** is a conceptual diagram illustrating a cross-sectional view of a folding device **100** with a flexible continuous display **106** and example supporting plates **140A** and **140B** (collectively, “support plates **140**”). Support plates **140** push up along the y-axis. FIG. **1C** may be a cross-sectional view of folding device **100** of FIG. **1A** along line A-A.

[0022] As previously noted, support plates **140** may be configured to support a backside of a continuous display, such as continuous display **106**. In some examples, support plates **140** may be collapsible support plates in that support plates **140** may fold away or otherwise be stowed when folding device **100** is closed. In some examples, actuation of support plates **140** may be performed by one or more components of the hinge assembly **122**.

[0023] In some examples, support plates **140** may be parallel with a plane of continuous display

106 when folding device **100** is open. For instance, as shown in the example of FIG. 1C, the upper surfaces of support plates **140** are parallel with a plane of continuous display **106** (illustrated as a dashed horizontal line). In some examples, the support plates **140** may accommodate deformations in continuous display **106** (e.g., the indent at the center). In other examples, supporting plates **140** may not be parallel with a plane of continuous display **106**, but may still accommodate deformations in continuous display **106**. In some examples, a gap may be present between supporting plates **140** to further help accommodate the deformations in continuous display **106**.

[0024] FIGS. 2A and 2B are conceptual diagrams illustrating example hinge assemblies **222** with supporting plates **240A** and **240B** (collectively, “support plates **240**”), in accordance with one or more aspects of this disclosure. Hinge assemblies **222** of FIGS. 2A and 2B may be an example of hinge assembly **122** of folding device **100** of FIGS. 1A and 1B. Similarly, support plates **240** may be an example of support plates **140** of FIG. 1C.

[0025] FIG. 2A is a direct perspective of a hinge assembly **222** of a folding device (e.g., folding device **100** of FIG. 1A-1C), in accordance with one or more aspects of this disclosure. FIG. 2B is an angled perspective of a hinge assembly **222** of a folding device (e.g., folding device **100** of FIG. 1A-1C), in accordance with one or more aspects of this disclosure. As shown in FIGS. 2A and 2B, hinge assemblies **222** may include support plates **240** comprising a single continuous plate folded to create two parallel plates, a first parallel plate **224A** and **224B** (collectively, “a first plate **224**”) and a second parallel plate **228A** and **228B** (collectively, “a second plate **228**”), assembly geometry **230**, and interflex connections **225**. Where the first plate **224** is closer to a rear of the continuous plate than the second plate **228**.

[0026] In some examples, interflex connections **225** may be one or more electrical connections (e.g., power and/or data) spanning the boundary between the two housings of the folding device. For instance, interflex connections **225** may include one or more ribbon cables routed between the two housings (e.g., housings **102** and **104** of FIG. 1C) across the hinge assembly **222**. Interflex connections **225** may be located below the second parallel plate **228**, proximal to the hinge assembly. In areas/regions of supporting plates **240** where the interflex connections **225** are located, weight reduction features are not present, and in some examples assembly geometry **230** is not present either. This is further shown in FIGS. 4A and 4B.

[0027] Assembly geometry **230**, as shown in FIGS. 2A and 2B may be a circular void designed as a hole for a screw. As one example, a void may be an absence of a portion of both the first plate **224** and second plate **228**. In another example, a void may be an absence of a portion of either the first plate **224** or the second plate **228**. In some examples, the screw in assembly geometry **230** may be used to join the first plate **224** and the second plate **228** together. As another example, the screw in assembly geometry **230** may be used to join support plates **240** to other components such as a scoop. In some examples the screw in assembly geometry **230** may be coplanar (i.e., flush, level, even, smooth, etc.) with the first plate **224**. As one example, a filler, padding or other object may be inserted in assembly geometry **230** instead of the screw, or in addition to the screw, and may be coplanar with the first plate **224**. In some examples, assembly geometry **230** is located between the fold in support plate **240** and the straight edge of the first plate **224**. Further, assembly geometry **230** may be located equidistant to the fold in the support plate **240** and the straight edge of the first plate **224**, closer to the fold in support plate **240**, or closer to the straight edge of the first plate **224**.

[0028] FIGS. 3A, 3B, and 3C are conceptual diagrams illustrating example supporting plates **340**, in accordance with one or more aspects of this disclosure. Support plate **340** may be an example of supporting plates **240A**, **240B**, **140A**, or **140B** of FIGS. 1C, 2A, and 2B. Similarly, a first plate **324**, a second plate **328**, assembly geometry **330A**, **330B**, **330C**, **330D**, and **330E**, and interflex connections **325** may be examples of a first plate **224**, a second plate **228**, assembly geometry **230**, and interflex connections **225** of FIGS. 2A and 2B. FIGS. 3A, 3B, and 3C illustrate support plate **340** isolated from the hinge assembly of FIGS. 2A and 2B. FIG. 3A shows the support plate **340** with the first plate **324** in primary view, FIG. 3B shows the support plate **340** with the second plate

328 in primary view, and FIG. 3C shows support plate **340** unfolded with both the first plate **324** and the second plate **328** in primary view.

[0029] In some examples, assembly geometry **330** may comprise one or more of **330A**, **330B**, **330C**, **330D** and/or **330E** (collectively, “assembly geometry **330**”). Assembly geometry **330** may comprise one or more voids. As noted above, a void may be an absence of a portion of the first plate **324** and the second plate **328** and may, in the instance of **330A** and **330B**, be a circular void designed as a hole for a screw. In one example, FIG. 3C shows assembly geometry **330A** on the first plate **324** and corresponding assembly geometry **330B** on the second plate **328**. In another example, assembly geometry **330B**, **330C**, **330D**, and **330E** are only present on the second plate **328** and are not configured to contain corresponding assembly geometry on the first plate **324**.

[0030] In some examples, assembly geometry **330** may consist of one or more modifications that include an addition to a portion of the second plate **328**. Further, in some examples, assembly geometry **330** may consist of modifications that include the addition to and absence of a portion of the second plate **328**. For example, assembly geometry **330C** and **330D** consist of modifications to the second plate **328** that include the creation of additions and voids.

[0031] In general, assembly geometry is primarily present on the second plate **328**. In some examples, where assembly geometry is present on the first plate **324**, it is less complex and still allows for a clean (i.e., smooth, flat, unblemished) first plate **324**.

[0032] FIGS. 4A and 4B are conceptual diagrams further illustrating example supporting plate **440**, in accordance with one or more aspects of this disclosure. Support plate **440** may be an example of support plate **340** of FIGS. 3A-3C. Similarly, a first plate **424**, a second plate **428**, assembly geometry **430A**, **430B**, **430C**, **430D**, and **430E**, and interflex connections **425** may be examples of a first plate **324**, a second plate **328**, assembly geometry **330A**, **330B**, **330C**, **330D**, and **330E**, and interflex connections **325** of FIGS. 3A-3C. FIGS. 4A and 4B illustrate support plate **440** configured with weight reduction geometry **432** and isolated from the hinge assemblies of FIGS. 2A and 2B. FIG. 4A shows the support plate **440** with the second plate **428** in primary view, and FIG. 4B shows support plate **440** unfolded with both the first plate **424** and the second plate **428** in primary view.

[0033] In some examples, weight reduction geometry **432** may comprise of one or more voids configured to reduce the weight of support plate **440**. In one example, weight reduction features **432** are only present on the second plate **428**, as illustrated by FIG. 4B. In another example, weight reduction features **432** may not be present in areas comprising the interflex connections **425**.

[0034] FIG. 5 is a conceptual diagram illustrating a cross-sectional view of example supporting plate **540**, in accordance with one or more aspects of this disclosure. FIG. 5 may be a cross-sectional view of folding device **100** of FIG. 1A along line A-A. Support plate **540** may be an example of support plate **440** of FIGS. 4A and 4B. Similarly, a first plate **524**, a second plate **528**, and assembly geometry **530C** and **530D** may be examples of a first plate **424**, a second plate **428**, and assembly geometry **430C** and **430D** of FIGS. 4A and 4B.

[0035] In some examples, support plate **540** may comprise a continuous plate folded over itself to create a fold **536** and a first plate **524**, parallel to, a second plate **528**. Where the fold **536** in the single continuous plate is opposite the straight edge **538** of the first plate **524**. In one example, the single continuous plate may be made from stainless steel and the first and second plates may be joined via spot welding. Such joining may provide added robustness/stiffness to the support plate **540** without weight penalty.

[0036] The following numbered examples may illustrate one or more aspects of this disclosure:

[0037] Example 1. A folding device comprising: a first housing; a second housing; a hinge assembly rotatably connected to the first housing and the second housing; a continuous display connected to and across the first housing and the second housing; and a first collapsible support plate connected to the first housing and configured to support the continuous display when the folding device is fully opened, wherein: the first collapsible support plate comprises a single

continuous plate folded over itself to form two parallel plates, a first parallel plate of the two parallel plates defines a straight edge that overhangs one or more voids in a second parallel plate of the two parallel plates, and the straight edge defined by the first parallel plate is adjacent to the hinge assembly.

[0038] Example 2. The folding device of example 1, wherein the first parallel plate of the two parallel plates comprises fewer voids than the second parallel plate of the two parallel plates and, wherein the voids in the second parallel plate are proximal to the hinge assembly.

[0039] Example 3. The folding device of example 1, wherein the voids in the two parallel plates comprise at least one or more of: assembly geometry; and weight reduction features.

[0040] Example 4. The folding device of example 3, wherein one or more interflex connections are routed between the first housing and the second housing, and wherein the weight reduction features are not present at regions of the single continuous plate where the interflex is routed.

[0041] Example 5. The folding device of example 3, wherein the assembly geometry further comprises a circular void configured to comprise a screw and/or a filler, wherein: a portion of the screw or the filler distal to the hinge assembly is flush with the first parallel plate of the two parallel plates, and the circular void is not intersected by the straight edge of the first parallel plate or the fold in the single continuous plate.

[0042] Example 6. The folding device of example 1, wherein a fold in the single continuous plate is opposite the straight edge of the first parallel plate.

[0043] Example 7. The folding device of example 1, wherein the first parallel plate and the second parallel plate of the two parallel plates are spot welded together.

[0044] Example 8. The folding device of example 1, wherein the single continuous plate is formed of stainless steel.

[0045] Example 9. The folding device of example 1, further comprising: a second collapsible support plate connected to the second housing and configured to support the continuous display when the folding device is fully opened, wherein, the second collapsible support plate comprises a single continuous plate folded over itself to form two parallel plates.

Claims

1. A folding device comprising: a first housing; a second housing; a hinge assembly rotatably connected to the first housing and the second housing; a continuous display connected to and across the first housing and the second housing; and a first collapsible support plate connected to the first housing and configured to support the continuous display when the folding device is fully opened, wherein: the first collapsible support plate comprises a single continuous plate folded over itself to form two parallel plates, a first parallel plate of the two parallel plates defines a straight edge that overhangs one or more voids in a second parallel plate of the two parallel plates, and the straight edge defined by the first parallel plate is adjacent to the hinge assembly.
2. The folding device of claim 1, wherein the first parallel plate of the two parallel plates comprises fewer voids than the second parallel plate of the two parallel plates and, wherein the voids in the second parallel plate are proximal to the hinge assembly.
3. The folding device of claim 1, wherein the voids in the two parallel plates comprise at least one or more of: assembly geometry; and weight reduction features.
4. The folding device of claim 3, wherein one or more interflex connections are routed between the first housing and the second housing, and wherein the weight reduction features are not present at regions of the single continuous plate where the interflex is routed.
5. The folding device of claim 3, wherein the assembly geometry further comprises a circular void configured to comprise a screw and/or a filler, wherein: a portion of the screw or the filler distal to the hinge assembly is flush with the first parallel plate of the two parallel plates, and the circular void is not intersected by the straight edge of the first parallel plate or the fold in the single

continuous plate.

6. The folding device of claim 1, wherein a fold in the single continuous plate is opposite the straight edge of the first parallel plate.

7. The folding device of claim 1, wherein the first parallel plate and the second parallel plate of the two parallel plates are spot welded together.

8. The folding device of claim 1, wherein the single continuous plate is formed of stainless steel.

9. The folding device of claim 1, further comprising: a second collapsible support plate connected to the second housing and configured to support the continuous display when the folding device is fully opened, wherein, the second collapsible support plate comprises a single continuous plate folded over itself to form two parallel plates.
